Test for sulphate/sulphuric acid:

Dilute solution of sulphuric acid/aqueous solution of sulphates gives white precipitate (barium sulphate) with barium chloride solution. It can also be detected using lead acetate solution. Here a white precipitate of lead sulphate is obtained.

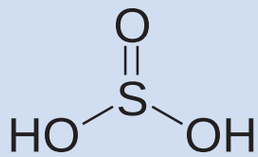
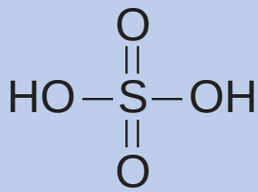
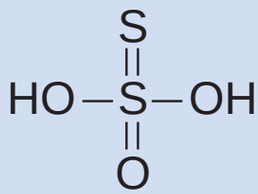
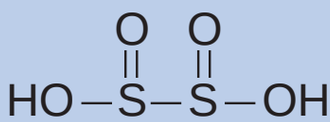
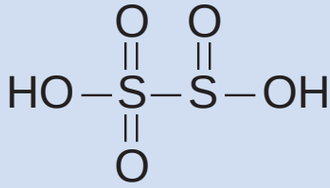
$$BaCl_2 + H_2SO_4 \longrightarrow \underset{\substack{\text{Barium sulphate}\\\text{(White precipitate)}}}{BaSO_4 \downarrow} + 2HCl$$

$$(CH_3COO)_2Pb + H_2SO_4 \longrightarrow \underset{\substack{\text{Lead sulphate}\\\text{(White precipitate)}}}{PbSO_4 \downarrow} + 2CH3COOH$$

Structure of oxoacids of sulphur:

Sulphur forms many oxoacids. The most important one is sulphuric acid. Some acids like sulphurous and dithionic acids are known in the form of their salts only since the free acids are unstable and cannot be isolated.

Various oxo acids of sulphur with their structures are given below

Name	Molecular Formula	Structure
Sulphurous acid	H_2SO_3	
Sulphuric acid	H_2SO_4	
Thiosulphuric acid	$H_2S_2O_3$	
Dithionous acid	$H_2S_2O_4$	
Dithionic acid	$H_2S_2O_5$	

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